

Comprehensive E-Commerce Business Analytics Report

Course End Project 1 — Operational Insights & Customer Segmentation

Project Overview & Scenario Context

Situation: Acting as a data analyst for an e-commerce company, this framework processes and organizes customer orders to unlock tactical business insights. The enterprise distributes an inventory matrix across channels including Electronics, Clothing, and Home Essentials (categorized as Home Goods within the automation pipeline). This comprehensive assessment maps frequent purchase selections, segments accounts based on total value metrics, and determines absolute profitability parameters per vertical. The analysis leverages native Python data structures (lists, tuples, dictionaries, and sets) paired with control structures to provide a data-driven infrastructure for inventory management and marketing deployment strategies.

Source Code Repository: github.com/chfreddy11/Course-end-project-1

1 & 2. Data Structure Storage & Category Mapping

- **Customer Master Registry:** Programmatically extracted into a primary tracking list of customer names.
- **Immutable Transaction Tuples:** Detailed order logs are preserved safely inside a global list containing structured (customer_name, product, price, category) tuples.
- **Account Purchase Index:** Modeled using a Python dictionary mapping customer keys directly to lists of ordered products.
- **Category Reference Dictionary:** Explicitly maps every dynamic item SKU directly back to its active category vertical.
- **Unique Categories Discovered:** Evaluated using set operations to display active channels: *Clothing, Electronics, Home Goods*.

3 & 5. Customer Spending & Tier Classification

A looping processing engine sums aggregate expenditures (V) per distinct customer ID. Account categories are then established through conditional structures according to the explicit ruleset thresholds: High-Value Buyer (>\$100), Moderate Buyer (\$50 - \$100), and Low-Value Buyer (<\$50). Adjustments to the product generator ensure balanced representation:

Customer ID	Total Cumulative Spend	Assigned Value Classification
CUST1001	\$1,410.20	High-Value Buyer
CUST1002	\$78.50	Moderate Buyer
CUST1003	\$24.15	Low-Value Buyer
CUST1004	\$1,120.60	High-Value Buyer
CUST1005	\$35.80	Low-Value Buyer
CUST1006	\$92.10	Moderate Buyer
CUST1007	\$845.30	High-Value Buyer
CUST1008	\$145.00	High-Value Buyer
CUST1009	\$41.90	Low-Value Buyer
CUST1010	\$1,350.15	High-Value Buyer

4. Core Business Intelligence Insights

- **Total Revenue per Vertical:** Dynamically accumulated via loops and saved into a category-revenue map.
- **Dynamic SKU Extraction:** Set comprehension isolates unique products across all historical customer purchases.
- **Targeted Electronics Filter:** Optimized list comprehension isolates exactly which unique customers purchased an product in the Electronics category.
- **Top Tier Consumer Sorting:** The system utilizes lambda sorting over total spends to flag the top three highest-spending customers for priority retention campaigns.

5. Behavioral Intersection Metrics (Set Operations)

- **Multi-Category Diversification:** Evaluated using set length evaluations (`len(categories) > 1`) to pinpoint omni-channel buyers who cross traditional vertical boundaries.
- **Cross-Over Cohort Detection:** Utilizes explicit set math intersections (`Electronics_Buyers.intersection(Clothing_Buyers)`) to track target customers buying both technology items and apparel.